

# Joohyun Lee

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## RESEARCH INTERESTS

Organic Electrochemical Devices (OECTs, OECDs), Conjugated Polymer Synthesis, Bioelectronic Applications

## EDUCATION

### Chung-Ang University (CAU)

- M.S. in Chemical Engineering  
GPA: 4.18 / 4.5
- B.S. in Chemical Engineering and Materials Science  
GPA: 3.75 / 4.5  
*Leave of absence for mandatory military service (May 2019–Dec. 2020)*

Seoul, Republic of Korea

*Sep. 2024–Present*

Advisor: Prof. Felix Sunjoo Kim

*Mar. 2018–Aug. 2024*

Advisor: Prof. Felix Sunjoo Kim

## PUBLICATIONS

1. “Morphology and Crystallinity of Transparent Capacitive Ion Storage Layer for Electrochromic Displays.”
  - Seungwon Lee, Yuseok Song, **Joohyun Lee**, Jaeyong Ahn, Won-June Lee, Jiao Suo, Yoonhoo Lee, Liyan You, Jianguo Mei, Felix Sunjoo Kim, and Inho Song\*.
  - *ACS Applied Materials & Interfaces*, accepted (2026).
2. “Enhancement-Mode PEDOT:PSS Electrochemical Transistors with Solid Electrolytes Based on [DBUH][OAc] as a Dual-Role Ionic Component.”
  - Donguk Kim<sup>†</sup>, **Joohyun Lee**<sup>†</sup>, Seungjin Song, Youngseok Kim, and Felix Sunjoo Kim\*.
  - Under review.
3. “Electrolyte-Driven Threshold Voltage Modulation in PEDOT:PSS OECTs via Anion Control of DBU-based Acid-Base Ionic Liquids.”
  - **Joohyun Lee**<sup>†</sup>, Donguk Kim<sup>†</sup>, Youngseok Kim, Seungjin Song, and Felix Sunjoo Kim\*.
  - Manuscript in preparation; submission expected in 2026.
4. “Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering and Oxidative Polymerization.”
  - **Joohyun Lee**, Seungjin Song, and Felix Sunjoo Kim\*.
  - In preparation.

## RESEARCH EXPERIENCE

### Prof. Felix Sunjoo Kim Lab, Chung-Ang University

*Graduate Researcher*

Seoul, Republic of Korea

*Sep. 2024–Present*

- **Project 1:** Electrolyte-Driven Threshold Voltage Modulation in PEDOT:PSS OECTs via Anion Control of DBU-based Acid-Base Ionic Liquids
  - Controlled the ionicity of DBU-based ABILs by systematically varying the pKa of carboxylic acid components used in synthesis, characterized via <sup>1</sup>H NMR spectroscopy and DFT calculations
  - Characterized threshold voltage shift in PEDOT:PSS OECTs as a function of ABIL ionicity-dependent de-doping level via chemical (UV-Vis-NIR, Raman, XPS), electrical (transfer/output curves), and electrochemical (CV, EIS) characterization
  - Implemented zero-*V*<sub>GS</sub>-load OECT inverters by exploiting the ABIL-tuned threshold voltage
- **Project 2:** Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering and Oxidative Polymerization
  - Synthesized EDOT-based D-A-D monomers via Stille coupling and removed organotin by-products by KF washing and column chromatography (confirmed by <sup>1</sup>H NMR)
  - Exploited the high reactivity of the α-site of EDOT for facile oxidative polymerization via aqueous-phase (PSS-templated) and surface methods, adding a base to retard the rapid polymerization during spin-coating
  - Determined the energy levels of synthesized monomers and polymers by correlating optical bandgap (UV-Vis-NIR) with electrochemical measurements (CV) and DFT calculations (ORCA)

### Prof. Felix Sunjoo Kim Lab, Chung-Ang University

*Undergraduate Researcher*

Seoul, Republic of Korea

*Mar. 2022–Aug. 2024*

- **Project 1:** Enhancement-Mode PEDOT:PSS Electrochemical Transistors with Solid Electrolytes Based on [DBUH][OAc] as a Dual-Role Ionic Component
  - Demonstrated that [DBUH][OAc] exhibits distinct interaction with PEDOT:PSS compared to [EMIM][TFSI], which is critical for enhancement-mode OECT operation
  - Revealed via DFT and XPS that [DBUH][OAc] preferentially interacts with the ion-conduction pathway in dry PEDOT:PSS, yielding C-V profiles with distinct accumulation and depletion regions analogous to organic semiconductors

- **Project 2: Long-Term Memory Organic Transistor Utilizing Electrolyte and Ferroelectric Polymer Bilayer Gate Dielectric**
  - Demonstrated synaptic memory with >24-hour retention and lower operating voltage in bilayer-gated organic transistors compared to ferroelectric polymer-only counterparts
  - Initially attributed the memory to ferroelectric polarization based on P-E hysteresis of the bilayer, but revealed through control experiments without the ferroelectric polymer that the long-term memory originated from the increased ionic resistance of the low-ion-concentration electrolyte (in collaboration with H. Yang)

## FELLOWSHIPS, AWARDS & HONORS

- **BK21 FOUR Fellowship**, Ministry of Education, Korea (monthly stipend, ~\$800) Sep. 2024–Aug. 2026
- **Research Assistantship**, CAU (full tuition waiver, ~\$6,000/semester) Fall 2025
- **Teaching Assistantship**, CAU Sep. 2024–Aug. 2025
- **Academic Excellence Scholarship**, CAU (ranked 1st in department) Spring 2024
- **Undergraduate Research Scholarship**, CAU (Practical & Research Engagement program; selected twice) Sep. 2022–Feb. 2024
- **Leadership Scholarship**, CAU (American Football Club president & captain) Mar. 2021–Feb. 2023

## SELECTED CONFERENCES

1. “Switching from Depletion- to Enhancement-mode in PEDOT:PSS-based Electrochemical Transistors.”  
Donguk Kim, **Joohyun Lee**, Jihoon Kim, and Felix Sunjoo Kim\*.  
*ITC 2024, 18th International Thin-Film Transistor Conference*, 2024.
2. “Threshold Voltage Modulation of PEDOT:PSS Based Electrolyte-Gated Transistors by Anion Control in Acid-Base Type Ionic Liquid.”  
**Joohyun Lee**, Donguk Kim, Youngseok Kim, and Felix Sunjoo Kim\*.  
*FPI 16 Symposium, 16th International Symposium on Functional  $\pi$ -Electron Systems*, 2025.
3. “Non-Volatile Switching Characteristics of Organic Electrochemical Transistors Enabled by Extremely Low Ion Concentration of Gel Electrolyte.”  
Heewon Yang, Jungmin Lee, **Joohyun Lee**, Minuk Kim, Hyejoon Park, Donguk Kim, Youngseok Kim\*, and Felix Sunjoo Kim\*.  
*FPI 16 Symposium, 16th International Symposium on Functional  $\pi$ -Electron Systems*, 2025.
4. “Toncity of Protic Ionic Liquid Influences Channel Conductance of Conducting Polymer in Electrochemical Transistors.”  
**Joohyun Lee**, Donguk Kim, and Felix Sunjoo Kim\*.  
*2026 KICHe Spring Meeting*, 2026.
5. “Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering.”  
**Joohyun Lee**, Seungjin Song, and Felix Sunjoo Kim\*.  
*KJF-ICOMEF 2026, 2026 KJF-International Conference on Organic Materials for Electronics and Photonics*, 2026.

## EXPERIENCE & ACTIVITIES

- **Teaching Assistant**, Department of Chemical Engineering, CAU Sep. 2024–Aug. 2025  
Special Topics in Chemical Engineering; Chemical Process Laboratory (process simulation with Aspen HYSYS)
- **Internship**, MeasHub Jul. 2026–Sep. 2026  
Electrical & electrochemical measurement platform where researchers share recipes and store results in the cloud. Supported program planning and coordination, and contributed to organizing and running conferences on site
- **President & Captain**, CAU American Football Club Mar. 2018–Feb. 2023  
Team member from 2018; president and captain (2021–2023), leading club operations and on-field performance. Seoul Collegiate American Football Tournament: Champion (2018), Runner-up (2019), 3rd Place (2021)

## SKILLS

- **Device & Electrochemical Characterization**: OECT fabrication and figure-of-merit extraction ( $g_m$ ,  $V_T$ , on/off,  $\mu C^*$ , transient), OECT inverters (gain, noise margin), CV, EIS, C–V ( $C^*$ ), spectroelectrochemistry [HP4156A, Ki2634B, SP-150e]
- **Synthesis**: Stille coupling, oxidative polymerization (solution & surface), protic ionic liquid synthesis, column chromatography
- **Structural & Spectroscopic**: XPS, Raman, UV-Vis-NIR, FT-IR, NMR
- **Morphological & Thermal**: AFM, TGA, DSC
- **Computational**: DFT/TD-DFT (ORCA, Spartan), Python, MATLAB, Aspen HYSYS
- **Languages**: Korean (Native), English (Proficient)

## REFERENCES

Dr. Felix Sunjoo Kim, Professor, Chung-Ang University  
 Dr. Inho Song, Assistant Professor, Chung-Ang University  
 Dr. Youngseok Kim, MeasHub (internship supervisor)

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